Gene Pulser® Electroprotocol

Cell Type Bacterial, gram positive

Species Used *Enterococcus hirae*

Molecules Electroporated DNA: linear; plasmids: 5 to 15 kb, supercoiled or relaxed.

Before the Pulse

Cell Growth Medium M-17 (very rich broth)

Growth Phase at Harvest O.D. (600) = late log

Pre-pulse Incubation none

Wash Solution water

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 25 °C

Electroporation Medium water

Cell Density 1- 2 x 10 (10) cells / ml

Volume of Cells 120 µl

DNA Concentration 1 pg to 1 µg

DNA Resuspension Buffer water

Volume of DNA 1 µl

Cuvette Gap 0.1 cm

Voltage 2.5 kV

Field Strength 25 kV/cm

Capacitor 25 µF

Resistor (Pulse Controller) See Comments

Time Constant 65 msec

After the Pulse

Outgrowth Medium M-17

Outgrowth Temperature 37 °C

Length of Incubation 30 min

Selection Method or Assay Used erythromycin resistance

Electroporation Efficiency 5 x 10 (6) transformants / µg DNA

Per Cent Survival 70%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Resistor Setting: 3000 Ω serial resistor
References: *Biochemie* **72**: 279-83 (1990),
TIBS 15: 175-77 (1990).

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Survey Number 061